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A systematic review of context-aware technology use in foreign language learning

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ABSTRACT

As context plays a critical role in foreign language (FL) learning, context-aware technology is a promising tool for supporting FL learning. The present study examined studies published between 2000 and 2018 to understand the trends of technology and related research and the current issues with context-aware technology in FL learning. This review included 75 empirical studies and 13 system-descriptive studies. The empirical studies were further examined in terms of publication year, methodology, participants, languages, device types, and the effectiveness of the technology based on FL learning results. Most of the studies indicated positive effects of the technology in the affective, linguistic, socio-cultural, and cognitive domains. The review also found issues with the reviewed studies, including the research methods, research sustainability, and practicability of the technology. The review results showed that the number of publications per year fluctuated rather than increased over the studied period. Students' perceptions of the use of context-aware technology in FL were the most common focus of the reviewed research; far fewer empirical studies were found that assessed students' actual language learning outcomes. The results of this study suggest that more long-term, pedagogy-informed empirical studies based on students' actual learning outcomes are required to verify the effectiveness of context-aware technology on FL learning and to help effectively integrate such technology into learning in the future.

KEYWORDS

Foreign language learning;
context-aware technology;
effectiveness; research trend

Introduction

Context has been greatly emphasized in learning in education (Collins, Brown, & Newman, 1988; Dewey, 1916; Lave & Wenger, 1991). According to Lave and Wenger (1991), cognition is socially defined and shaped, bound to the community of practice and constrained by social context, all of which aid cognition. Lave and Wenger further stated that 'learning is an integral part of generative social practice in the lived-in

world' (p. 35). From this perspective, knowledge does not reside in individuals' minds but is constantly emergent, constructed, interpreted, negotiated, and distributed among the members of a community. Hence, learning should be situated in context because it is a process of enculturation through which novice members learn to become full members of a community of practice by participating in activities. However, the current classroom situation often separates learning from the contexts in which knowledge is constructed and used. This situation has numerous disadvantages, including abstracting knowledge, demotivating students, and making learning inauthentic and unmeaningful (Collins et al., 1988). Context has an even more important role in the language learning process. Because language facility is an outcome of social interaction within a discourse community, it is inseparable from its socio-cultural context. However, foreign language (FL) learning frequently occurs in classrooms in a decontextualized manner (Lee & Park, 2019; Yang, 2006). Over the past several decades, huge efforts have been expended to contextualize FL learning activities (i.e. simulations, role-playing, or utilizing videos) and to connect learners with the target culture. These efforts, however, often have limitations, such as pre-fabricated scenarios or teacher-manipulated activities (Blyth, 2018; Liu & Tsai, 2013). FL learning is difficult to contextualize because the learning site is remote from the socio-cultural and discourse community of the language.

Advancements in context-aware technology have opened up new possibilities for connecting learning to its context (Lee & Park, 2019; Shadiev, Hwang, & Huang, 2017). Context-aware technology, which can detect the locations of users, can offer information more appropriate to users' needs and current surroundings. In particular, various built-in location-tracking features in mobile devices have made context-aware technology possible, which in turn supports situated learning. This technology, through social connectivity and global networks, connects learners to other users and allows them to share information both locally and worldwide (Godwin-Jones, 2017). Context-aware technology has been tested and implemented in various learning fields, and its educational benefits have been reported from diverse perspectives. From an affective perspective, context-aware technology increases learner motivation, engagement, interest, satisfaction, and confidence by engaging learners in real-life situations (Edmonds & Smith, 2017; Hemmi, Narumi-Munro, Alexander, Parker, & Yamauchi, 2014; Hwang & Wu, 2014) which, in turn, leads to active participation and meaningful learning. From a cognitive perspective, context-aware technology helps promote procedural knowledge and enhances the understanding of abstract concepts by visualizing them (Dede, 2009; Kerawalla, Luckin, Seljeflot, & Woolard, 2006).

Context-aware technology also facilitates meta-cognitive skills, such as autonomous learning and self-efficacy (Hwang, Shi, & Chu, 2011; Sun, Chang, & Chen, 2015). Regarding language learning, context-aware technology enables situated language learning, in which learners can experience authentic language input and generate creative language output (Cheng, Hwang, Wu, Shadiev, & Xie, 2010; Wong, 2013). Moreover, this technology enriches social interactions, which are essential to language learning (Lee & Park, 2019; Thorne, Hellermann, Jones, & Lester, 2015) and enhances learning efficiency by connecting formal learning to informal and non-formal learning (Shadiev et al., 2017). Due to its potential benefits to learning, it is expected that context-aware technology will become more widely used in FL learning in the near future. Since the advent of context-aware technology in the early 2000s, considerable related research has been published. Therefore, examining the trends and the effectiveness of context-aware technology on FL learning in this existing body of research is beneficial.

Literature review

FL learning and context

Context plays a key role in learning because knowledge is inseparable from its context and from the activities by which it is acquired (Collins et al., 1988; Dewey, 1916). Meaningful learning occurs only when learning is embedded in a context. Context, simply put, can be anything ‘that surrounds people who can listen [to], see, and contact others to interact’ (Chen, Chang, Lin, & Yu, 2009). In other words, context includes temporal, spatial, social, environmental, and material information. Dewey (1916) particularly emphasized the role of social contexts as well as the physical contexts in which learning takes place. A substantial portion of the literature has presented the educational benefits of contextualized and situated learning. By placing learners in real-life situations, situated learning enhances their understanding of knowledge; therefore, it reduces learning time and enhances learning efficiency (Chen & Li, 2010). Situated learning also helps learners transfer and apply newly acquired concepts and knowledge to novel situations. Situating learners in authentic problem-solving situations enhances learners’ motivation and engagement, actively involves them in the activity, and prevents inert knowledge (Collins et al., 1988; Liu, 2009). In addition, situated learning promotes collaboration and interaction (Ozudogru & Ozudogru, 2017; Schell & Black, 1997). Overall, situated learning facilitates meaningful learning and increases learning effectiveness.

Context is even more important in language learning because language is largely context-dependent, context-sensitive, and closely bound to the situation in which it is used, including the participants, the relations among the participants, and societal rules. From a social constructivist perspective, learning a language involves becoming a functional member of the discourse community. First language (L1) acquisition is a perfect example of situated learning. A first language is naturally acquired through interactions with other members of the community and by participating in activities in diverse social-cultural contexts. A newcomer, initially a peripheral participatory member, later becomes a fully functional member of the language community.

However, the FL learning classroom is often contrasted with situated learning because FL learning classrooms remove learning from context and present knowledge in a decontextualized manner (Ahn & Lee, 2016; Lee, 2017; Yang, 2006). As a result, a dichotomy is created between real-world learning and classroom-based learning; the latter makes language knowledge abstract and indirect and separates learners from the original value, practical utility, and relevance of the knowledge to the learner (Lunce, 2006). Thus, FL learning becomes a ‘second-hand experience’ to many learners. This, in turn, makes learning more difficult and reduces learners’ intrinsic motivation and interest in learning. Therefore, it is imperative to re-connect the FL classroom to the real world.

FL learning with context-aware technologies

Context-aware mobile technology has opened up new possibilities for situated and contextualized learning (Cochrane & Farley, 2017). The term ‘context-aware technology’ refers to a system that can detect a user’s location and gather information about the surrounding environment and the objects around the user using positioning techniques, QR codes, RFID, and similar technologies. Such systems provide unique technical affordances, such as portability, social interactivity, context sensitivity, connectivity, and individuality (Squire & Klopfer, 2007). According to Reinders and Pegrum (2016), all these features support learning. Portability allows learners to move around between places and between formal and informal settings, making ubiquitous learning possible. Social interactivity promotes interaction and collaborative learning, which is essential to language learning. Context sensitivity enables situated learning. Connectivity connects learners to other learners, teachers, and resources, enabling scaffolding and just-in-time support. Finally, individuality caters to individuals’ needs and facilitates personalized learning. Context-aware technology is usually combined with augmented reality

(AR) technology, which can double the learning effect. AR is “a situation in which a real world context is dynamically overlaid with coherent location or context sensitive virtual information” (Klopfer & Squire, 2008, p.205). AR constructs a blended reality consisting of both real and virtual worlds, offering learners ubiquitous learning opportunities in authentic environments and an immersive experience (Dede, 2011).

The terms ‘context-aware’ and ‘location-/place-based’ have been interchangeably used in several studies. However, there is a slight difference between the terms. Location-/place-based technology connects the user with the information about the specific, real, and physical place where he/she is positioned with the mobile device, thus providing somewhat limited resources. On the contrary, context-aware technology can be both location-dependent and location-independent. It provides information ‘that can be used to characterize the situation of an entity such as a person, place, or object that is considered relevant to the interaction between a user and an app’ (Nguyen, Pham, & Ho, 2010, p. 53). In other words, context-aware technology provides information about location, time, goals, resources, and the personal and social activities of individuals and groups more extensively.

Prior studies have examined the effectiveness of context-aware technology in language learning and reported positive outcomes overall. The most significant advantage of the technology is that it can connect learners with real-life situations beyond the classroom, such as workplaces, museums, or campuses, where they can learn and use the target language, thus making situated language learning possible. From an affective perspective, context-aware technology can increase learners’ motivation, interest, and engagement by involving them in authentic activities in real-life situations. Some studies have reported that simply conducting an activity outside the classroom increased the fun and enjoyability of learning (Hwang, Chen, Shadiev, Huang, & Chen, 2014; Lee & Park, 2019). Hwang et al. (2014) reported that learning with context-aware technology improved English-learning outcomes by reducing learners’ anxiety and increasing their motivation. From a social perspective, context-aware technology facilitates interaction and collaboration among participants. Through on-campus empirical investigations of micro-interactional and linguistic practices, Thorne and Hellermann (2017) found that devices with place-based technology generated a rich environment for interactions and mediated complex interactions among participants.

From a language learning perspective, educational benefits have been reported with regard to diverse language areas and skills. In one of the early studies on vocabulary learning with context-aware technology, Ogata and Yano (2004) proposed the use of Tag Added Learning Objects

(TANGO), which allowed students to learn vocabulary related to the objects surrounding them by utilizing mobile phones and RFID. Edge, Searle, Chiu, Zhao, and Landay (2011) developed MicroMandarin, a smartphone-based vocabulary learning context-aware application with a global positioning system (GPS) and investigated students' perceptions of learning using that application. The results showed that a majority of the students were satisfied with both the application and the content. Similarly, Dearman and Truong (2011) evaluated students using Vocabulary Wallpaper, a prototype casual Italian-learning application, and found that students acquired more context-dependent vocabulary than context-independent vocabulary. Chen and Li (2010) also reported that an experimental group using context-aware technology outperformed a control group in vocabulary learning because the technology afforded the opportunity to provide the 'right content' at the 'right time' and the 'right place.'

In terms of oral language skills, Liu, Tan, and Chu (2010) claimed that students with HELLO, QR code and AR-supported mobile English learning systems, had increased motivation; these systems effectively assisted their learning and enhanced their listening ability. Cheng et al. (2010) showed that students with context-aware technology demonstrated greater willingness to speak and performed better at English presentation. They further explained that environmental familiarity (campus) eliminated language barriers, reduced language anxiety, and positively affected learning outcomes. Holden and Sykes (2011) and Ogata et al. (2008) also showed how context-aware applications, Mentira (a community-based Spanish learning app) and LOCH (a Japanese-learning outside-the-classroom app) could promote students' communication skills by actively involving them in the community and in real-life situations.

Only a few studies have investigated writing with context-aware technology. The studies of Cheng et al. (2010) and Liu and Tsai (2013) revealed that when students were equipped with context-aware technology during writing tasks, they produced better-quality writing that featured greater linguistic complexity and more meaningful essays. The studies of Martin and Ertzberger (2013), Wu, Sung, Huang, Yang, and Yang (2011), and Liu, Kuo, Shi, and Chen (2015) reported that context-aware technology promoted FL reading skills and provided students with a sense of satisfaction and usefulness. Reinders, Lakarnchua, and Pegrum (2014), however, presented a somewhat contradictory result. In their study, while the experimental group using a location-based app produced more detailed descriptive language after a campus tour, the control group in the traditional classroom setting produced more syntactically complex language.

While it is controversial whether pragmatic knowledge is teachable, some studies have focused on the acquisition of pragmatics. One good example is Mentira, a game-based approach that provided effective and meaningful means for teaching pragmatics by engaging students in authentic communication with local people. Yin, Ogata, Tabata, and Yano (2010) utilized a context-aware app to support the acquisition of Japanese politeness by assigning different roles to the students. Considering that pragmatics is highly dependent on the context at hand and the individual participants (Godwin-Jones, 2016), it is difficult to acquire pragmatics in the classroom outside of the context of their use. Overall, context-aware technology, which situates learners in real-world situations and allows them to interact with real people in communities, effectively promotes learners' pragmatic knowledge and behaviours. Similar to pragmatics, culture is another area that is difficult to teach in FL learning. Culture cannot be fully taught in the classroom; it is experienced only by participating in the culture, so context is critical to learning a target culture. Holden and Sykes (2011), Ogata et al. (2008), and Shih, Chu, Hwang, and Kinshuk (2009) argued that context-aware technology promotes the acquisition of cultural knowledge by immersing learners in the target culture and enabling interactions with locals.

Most prior studies on FL learning with context-aware technology have reported educational benefits from diverse perspectives, although a few have presented mixed results concerning the effectiveness of the technology in FL learning. However, because the technology is still in a rudimentary stage, insufficient evidence-based empirical research has been conducted in this field. Moreover, no meta-analysis or review study has been published regarding the use of context-aware technology on language learning. This paper addresses the following questions:

1. What were the trends in context-aware technology in FL learning between 2000 and 2018?
2. What research methodologies were used in the reviewed studies?
3. What were the research topics and focus of the reviewed studies?
4. What were the research outcomes regarding context-aware technology in FL learning?
5. What issues and concerns are implied by the reviewed studies?

Methodology

For researchers and practitioners, it is crucial to know whether a certain pedagogical tool or teaching method is effective for student learning before implementing it into their classrooms. However, it requires a

considerable amount of time and efforts to search and review each individual literature relevant to the issue and determine whether to use it or not. On the other hand, a well-designed meta-analysis and a systematic review can save their time by providing pre-filtered evidence and potentially high levels of internal validity of evidence regarding the issue of interest. A meta-analysis and systematic review aims to synthesize the results of multiple original studies by employing scientific and systematic methods that delimit bias and errors (Petticrew & Roberts, 2006). Since there were not enough quantitative studies found to calculate an effect size and statistically measure the effectiveness for a meta-analysis, the current study employed a systematic review to provide evidence on the effectiveness of using context-aware technologies on FL learning to researchers and practitioners in the field.

The data for the current study were collected from major academic journal databases, such as ProQuest, Science Direct, EBSCO, and Cambridge Core, and from Google Scholar using the following keywords: 'context-aware,' 'location-based,' 'place-based,' 'ubiquitous,' 'language learning,' 'foreign language learning,' and 'second language learning.' The search included only articles published since 2000. In addition, a manual search of major journals related to technology and language learning was conducted to obtain additional articles; these journals included *CALL*, *ReCALL*, *Language Learning and Technology*, *BJET*, *Digital Creativity*, *Computers and Education*, *AJET*, *System*, *CALICO Journal*, and *Educational Technology and Society*. The studies that examined ubiquitous learning without actually involving context-aware technology were removed from the first round of the search results, resulting in 118 studies. After reviewing all initial 118 studies, 26 were eliminated because they were merely conceptual or theoretical papers about context-aware technology and did not involve the actual development of a system or the use of context-aware technology in FL learning. Additionally, some of the search results were published both as conference proceedings and journal articles (multiple publications, $N=3$); consequently, only one – usually the article, which typically included richer empirical data – was adopted. At the final stage of data collection, 88 studies were selected for inclusion in the current study.

The current study employed multiple analysis steps. First, the collected studies were divided into two groups: empirical studies ($N=75$) and non-empirical studies ($N=13$). To investigate the effectiveness of context-aware technology in FL learning, only empirical studies should be included; however, non-empirical studies that described cutting-edge context-aware technology systems and applications for language learning based on actual development experiences were expected to show trends

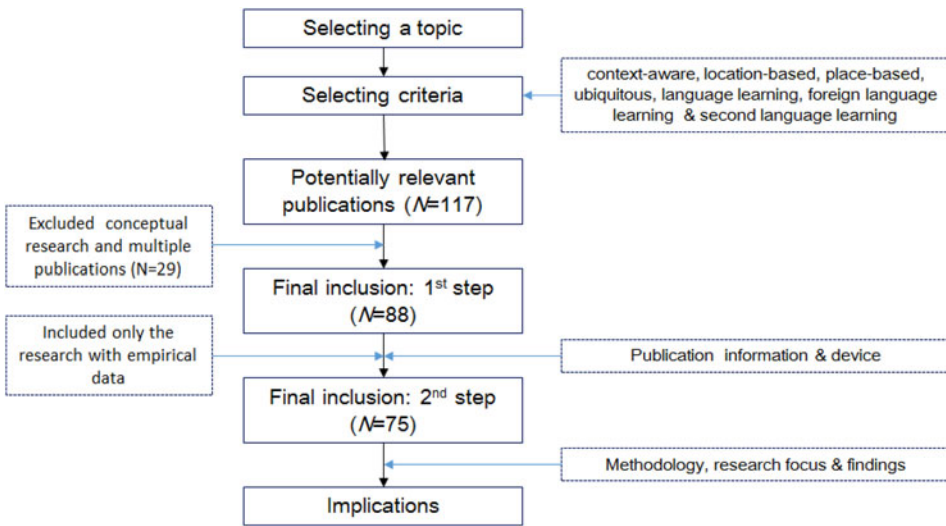


Figure 1. Research methodology.

in this field. Thus, those studies were also included during the first analysis step. In the second step, detailed information about each study was recorded in an Excel file, including publication information, technology type, topic, research focus, research method, L1 and target language (TL), information about the participants, the language skills included, and the research outcomes. Non-empirical studies were included only for publication information, the device used, and language as these studies did not report empirical data. In the third step, the frequencies of each type of information were calculated to identify trends. Finally, the outcomes of the studies were examined to investigate the effectiveness of context-aware technology in FL learning; then, the studies were categorized into positive/negative outcomes and different domains. The procedure of the research methodology was summarized in [Figure 1](#).

Results

Publication information and devices used

The first publication on context-aware technology and FL education appeared in 2004; however, most of the research in this field was published after 2010. Publications before 2010 are scarce because context-aware technology was still in a very early stage and thus inapplicable to FL education. The result showed that most of the research was published between 2010 and 2016 ([Figure 2](#)). The majority of studies were articles ($N=65$), followed by proceedings ($N=17$) and book chapters ($N=6$). Of 88 studies, 82 employed self-developed applications, while only six utilized existing applications. Several studies utilized basically the same self-

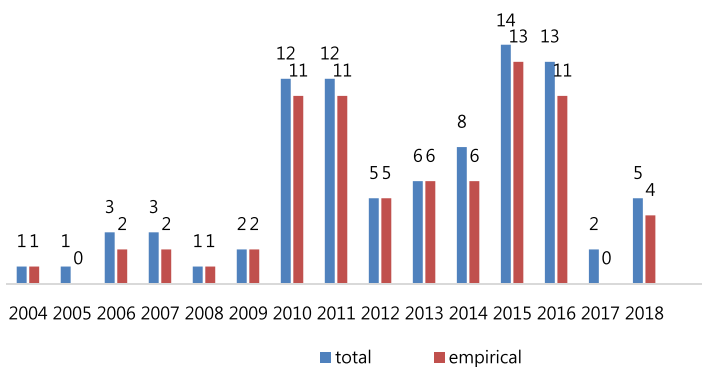


Figure 2. Publication year.

developed applications; however, with regard to research, they ranged from small pilot tests to larger experimental tests, while technologically, they modified or added new functions to the previous model in the subsequent research. A few studies reported the same type of case study using different languages. In addition, most of the studies utilized mobile phones ($N=67$) rather than PDAs ($N=20$) or tablets ($N=1$), and all of the studies employed context-aware technology such as GPS sensors, RFID, QR codes, WLANs, or other network positioning techniques.

Regarding language, Chinese ($N=30$) was the most common L1 found in the studies, followed by English ($N=8$) and Japanese ($N=5$) (Figure 3). The use of various L1s also appeared in approximately one-third of the studies. In terms of the TL, English was the majority language ($N=59$). The most common L1-TL pair was Chinese-English ($N=30$). This result seems reasonable considering the large populations of Chinese speakers (L1) with English as the TL.

Methodology

In this section, only the 75 studies with empirical data are analysed. Among the 88 selected studies, the 13 that focused only on describing self-developed systems without reporting on their actual use in FL learning practices or empirical data are not discussed in this section.

In terms of research methodology, both quantitative ($N=32$) and mixed research methods ($N=29$) were frequently employed (Figure 4). The descriptive studies ($N=6$) briefly described students' experiences and perceptions of the technology based on observations, casual interviews, or informal feedback from students without in-depth data collection or analysis. Because the majority of studies explored learners' perceptions, surveys were the most common data collection method.

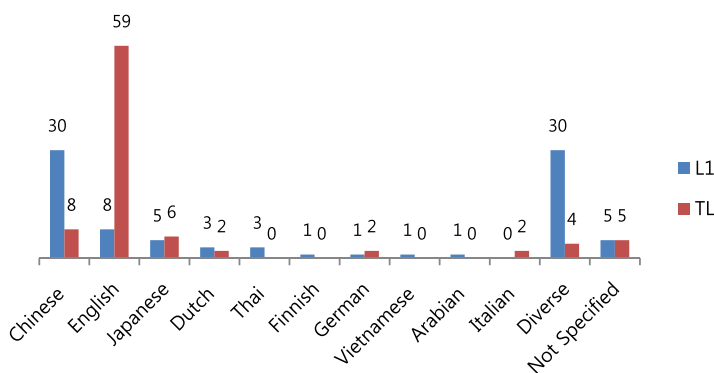


Figure 3. L1 and TL prevalence.

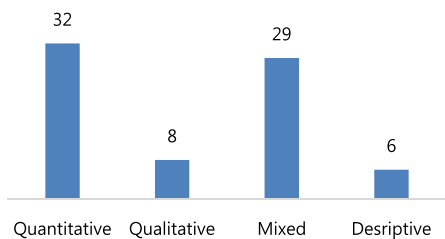


Figure 4. Research methodology.

Accordingly, descriptive statistics were the most frequently employed data analysis method.

The number of research participants varied widely, from one to 116. As shown in Figure 5, a significant number of studies involved fewer than 10 participants ($N=11$), and several studies did not report the number of participants ($N=6$) because those studies focused more on description and exploration of the newly developed system and less on learner experiments. As such, in terms of the research period, the number of one-time sessions and the number of studies with not-specified sessions were also relatively high. These studies employed observations, brief surveys or short interviews to validate system usability, usually following a lengthy description of the employed system (Figure 6).

The nationalities of the participants also varied (Figure 7). A total of 24 studies involved Taiwanese participants, which indirectly implies that at least 24 of the studies included in the current study were conducted by Taiwanese researchers or in Taiwan. In 19 studies, participants were from diverse countries rather than from a single country. Regarding educational levels, most of the studies employed college students ($N=45$), but 10 worked with elementary school students, 6 with secondary school students, and 4 with diverse participant educational levels (Figure 8). The overwhelmingly high number of college-level participants is assumed to be

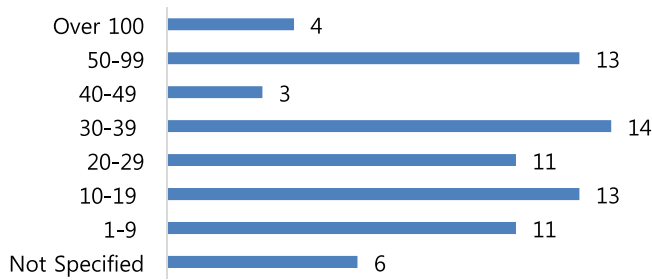


Figure 5. Number of participants.

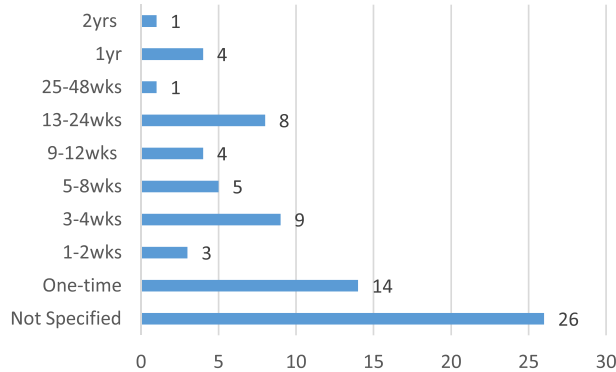


Figure 6. Research period.

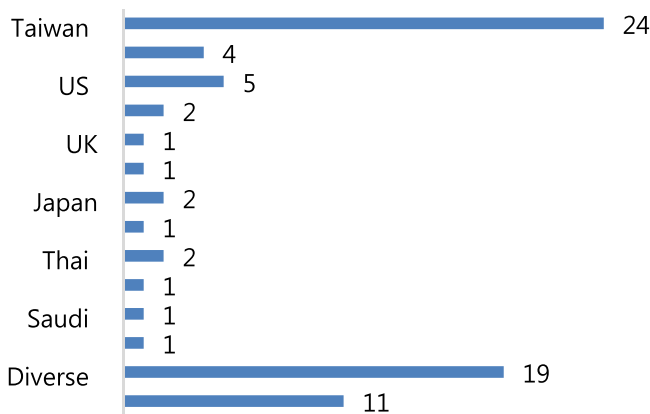


Figure 7. Nationalities of participants.

ascribable to the fact that the researchers employed students attending their institutes and utilized the campus as an easily accessible location.

The experiments were conducted in a variety of places; however, more than half of the experiments were conducted on college campuses or at schools ($N=42$), and 10 were conducted in a city (Figure 9). Some studies utilized specific sites, such as a restaurant ($N=1$), kitchen ($N=1$), workplace ($N=1$), cinema ($N=1$), fitness centre ($N=1$), home ($N=1$),

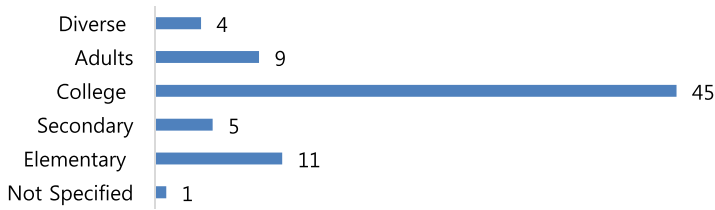


Figure 8. Educational level of participants.

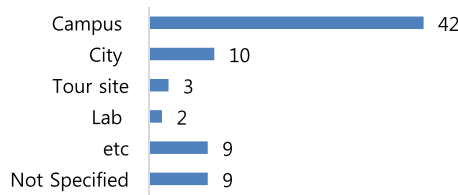


Figure 9. Experiment locations.

zoo ($N=1$), or shopping centre ($N=1$). The other two studies, which were quasi-experimental studies on vocabulary learning with context-aware technology, were conducted in a lab in which various objects were tagged with RFID. Eleven studies did not specify the location where the experiment occurred.

Research focus

The most frequently appearing research focus was learner perceptions ($N=70$). Because most of the studies involved the development of new systems or applications, they tended to evaluate the effects of technology on FL learning by investigating learner perceptions of playfulness, motivation, and the technology acceptance model (i.e. usefulness and ease of use) and language learning. These studies examined the effectiveness of the systems on language learning through surveys, interviews, and feedback from the participants. The remainder of the studies focused on system accuracy ($N=2$), interactions ($N=1$), and language proficiency ($N=2$). Of the 75 experimental studies, only 17 reported students' actual learning outcomes and measured them through the experimental research design. Fifteen studies investigated both students' perceptions and their language learning outcomes.

The current study further investigated the domains with which each study was most concerned: affective, linguistic, cognitive, metacognitive, socio-cultural or system (Figure 10). The results showed that the majority of studies focused on the affective ($N=43$) and linguistic domains ($N=61$). The affective domain encompasses enjoyability, engagement, interest, motivation, confidence, and satisfaction when using the system.

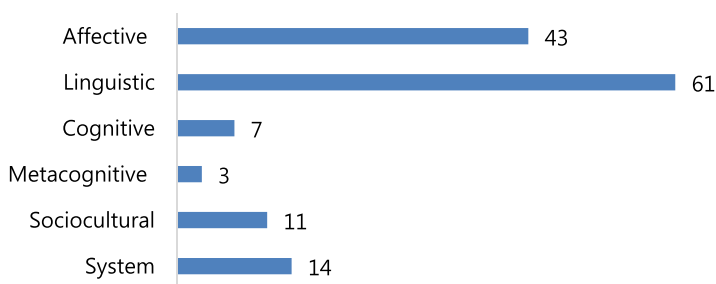


Figure 10. Research focuses.

The studies which focused on the linguistic domain examined how the context-aware technology facilitated students' TL learning. A small number of studies were concerned with students' cognition and metacognition. In the cognitive domain, research investigated memorization and recall in vocabulary learning ($N=7$). Studies on the metacognitive domain examined learner autonomy, efficacy, and regulation ($N=3$). Eleven studies focused on the social domain, examining collaborative learning, interactions, and cultural learning. Additionally, 14 studies described the system through brief observations or data from short interviews. In [Figure 10](#), the total number of domains ($N=139$) is larger than the total number of studies ($N=75$) because some studies addressed more than one domain. For instance, [Chen and Li \(2010\)](#) investigated students' perceptions of system usability and also measured their vocabulary learning outcomes. The outcomes of the reviewed studies will be further discussed in the following section.

The present study further examined 61 studies in the linguistic domain and categorized them based on language areas and skills. While vocabulary was most commonly employed in the system and explored in research ($N=29$), other skills were under-researched. Thirteen studies explored more than one language skill. Fourteen studies, which mostly focused on testing system usability and accuracy, did not specify language skills. The language areas and skills investigated in the studies are summarized in [Figure 11](#).

Findings of the reviewed studies

Most of the reviewed studies reported positive results in diverse areas of learning. From an affective perspective, the majority of studies maintained that context-aware technology increased students' positive emotions towards language learning. In total, 43 studies in the affective domain reported increased motivation, enjoyability, engagement, flow, fun, and excitement and reduced anxiety during language learning activities, and 11 studies among them additionally reported increased

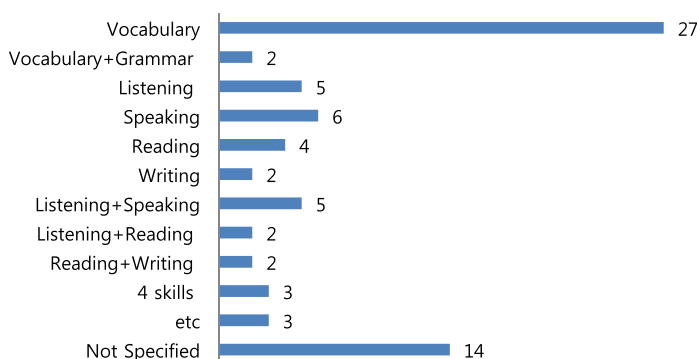


Figure 11. Language areas and skills.

satisfaction and confidence. The studies also implied that the most significant reason underlying the effectiveness of context-aware technology for language learning was closely related to the affective domain. The majority of the reviewed studies that examined students' language learning outcomes also examined their affective factors. The results showed that students' increased motivation, engagement and enjoyability during learning led them to participate more actively in the activities and spend more time learning, which ultimately resulted in enhanced learning outcomes (Harley, Poitras, Jarrell, Duffy, & Lajoie, 2016; Hwang et al., 2014; Reinders, Lakarnchua, & Pegrum, 2014). The studies dedicated to examining system usability and interfaces ($N=14$) showed that participants were satisfied with the system and found it useful to learning overall. These studies usually described the system under study in full detail, including a brief analysis of learners' perceptions of system usability.

In the cognitive domain, all seven studies claimed that using context-aware technology promoted students' understanding, memorization, recall, and retention rates in language learning, particularly in vocabulary learning. From a metacognitive perspective, three studies found that context-aware technology promoted autonomous learning. In terms of self-regulation and self-efficacy, one study argued that the technology helped to enhance self-regulated learning and self-efficacy, while the other study did not find a significant difference in pre- and post-test scores. Regarding the socio-cultural domain, all 11 reviewed studies found that context-aware technology facilitated interaction, collaborative learning, community learning, and cultural learning. Particularly four studies confirmed that context-aware technology, by immersing students in authentic social conversations, facilitated pragmatic competency and richer use of communicative strategies.

In total, 61 studies investigated the effectiveness and usefulness of context-aware technology in the linguistic domain and 41 reported positive

results, while two reported partially positive results. In terms of language areas and skills, 29 studies explored the effectiveness of the technology for vocabulary learning. These studies reported that context-aware technology promoted vocabulary learning for the following reasons: (1) the technology detected students' surroundings and assisted vocabulary learning through adaptive technology, (2) the real-world relevance of the words based on actual matches between words and objects fostered students' understanding and memorization of the words, and (3) the technology offered meaningful interactions with others through which the students were able to use the acquired vocabulary.

The studies showed that technology also enhanced listening, speaking, reading, and writing skills because (1) they were more motivated toward learning with the context-aware app, (2) context-familiarity reduced language anxiety and facilitated language learning, and (3) students acquired the language skills in meaningful contexts where they could connect what they learned to the way skills are applied in real communication situations, unlike in classroom learning, which usually lacks context. Particularly in speaking and writing, context functioned as a good springboard to enhance learning. The study by Hwang et al. (2014) reported that the familiar context inspired students to write more and produce better-quality writing with higher levels of linguistic complexity, reasoning, and organization. Unlike other studies, Reinders, Lakarnchua, and Pegrum (2014) found only partially positive results of the technology on writing. Based on their results, they argued that some features in writing, such as discourse markers, are better learned in class, while descriptive skills are better acquired using the technology. They further found that students' enthusiasm and interest during the tour-creating activity diminished during the tour-taking activity. Similarly, studies reported that when students had greater flexibility to explore locations based on their interests and needs and were given opportunities to create their own content rather than merely following a ready-made system path, they found learning more enjoyable and meaningful (Liu & Chu, 2010; Reinders & Lakarnchua, 2014; Shih et al., 2009; Wong, 2013).

While the reviewed studies presented the educational benefits of using the context-aware technology in detail, the disadvantages were mentioned only briefly. The disadvantages included small screen sizes, slow Wi-Fi speeds, inaccurate location detections, and the large storage space required by the application. In addition, although one study mentioned that a PDA was more effective for learning than a mobile phone due to the larger screen size of the PDA, in general, the studies did not report on the advantages of one technology over another.

Discussion

Herrington, Oliver, and Reeves (2003) claimed that situated learning environments should provide authentic context, authentic activities, access to expert performances, and multiple perspectives and coaching, support the collaborative construction of knowledge, and prompt reflection and articulation to enable tacit knowledge to be made explicit. The reviewed studies confirmed that context-aware technology provides these authentic context conditions. Ubiquitous learning in the past allowed only unidirectional access from the user to the device. In contrast, ubiquitous learning with context-aware technology enables bidirectional or multidirectional access. The context-aware technology can detect individual learners, collect data such as learner profiles, locations, other users, and ambient information, offer individualized guidance and personalized learning, and facilitate learning in context (Holden & Sykes, 2011). Specifically, the advancement of iPhone and Android smartphones since 2007 has enabled context-aware technology that, according to Godwin-Jones (2017), offers the most powerful affordance for situated learning. Learning FL in an authentic, real-life situation apparently makes learning more fun and enjoyable, thereby increasing students' intrinsic motivation and engagement and helping them participate more actively in learning. Most of the reviewed studies indicated that context-aware technology is beneficial to student FL learning from affective, cognitive, linguistic, and socio-cultural perspectives and indicated that context-aware technology facilitated FL learning. However, the present study also found important issues in the reviewed studies regarding research methods, research sustainability, and the practicality of both the research and the technology.

Research methods

The majority of the reviewed studies were exploratory studies, which often involved a pilot test on the system and quasi-experiments on students' perceptions of the usability and usefulness of the system in FL learning. The current study found that 82 of 88 studies had developed their own application or system for FL learning. Considering that most of the studies employed self-developed software, it was not surprising that many of them were exploratory or quasi-experimental studies that intended to test the feasibility of their systems on FL learning. These studies often involved a small number of participants and a short research period, and many did not report information about either the participants or the research period. Moreover, some studies conducted one-time session research with one participant. Of 75 experimental studies, only 17 studies directly measured students' language learning

outcomes. While the studies with empirical data on language learning outcomes tended to have a larger number of participants and longer research periods, the exploratory studies usually relied on students' perceptions of their language learning outcomes after participating in the experiment for only a short period. As shown in the studies by Reinders and Lakarnchua (2014) and Lee and Park (2019), a small change in learning conditions may affect students' attitudes and behaviours. In addition, students' initial enthusiasm towards learning with context-aware technology may decrease over time. Alternatively, some technical problems may appear during later stages, such as limited time and space for data storage, delayed update, or incompatibility with the updated operating system (OS). Hence, longer-term research may reveal a different picture of the usefulness and effectiveness of context-aware technology in FL learning on students' emotions, behaviours and language learning or reveal concerns about the technology itself.

The results of this study showed that most of the learning environments in the studies were at sites familiar to the students. While unusual locations such as tourist sites or zoos can attract students' attention, from a language learning perspective, these may not be as useful as familiar sites where students conduct daily activities. Familiarity with an authentic learning environment is important in FL learning because students can better utilize their previous knowledge and apply the target language to real-life situations in a familiar environment. Shadiev et al. (2017) also claimed that a familiar environment reduced students' language anxiety. While utilizing familiar sites in such studies may be beneficial to students, future research should expand sites beyond the campus to include additional contexts in which students' daily activities occur. Additionally, the reviewed studies were skewed in terms of nationality, locations, language skills, and participant populations; thus, studies with more diversity are required in future research.

Because context-aware technology is still in its rudimentary stage in education, the advent of self-developed applications for research and in educational practice is encouraging. However, more evidence-based, long-term empirical studies are needed to ensure the effectiveness of context-aware technology in FL learning in the future. Long-term research is also necessary to assess the sustainability of research in this field, which will be discussed in the following section.

Research sustainability

The publication years of the reviewed studies revealed interesting patterns: research on context-aware technology in FL learning was concentrated in

certain years (2010, 2011, 2015, and 2016) rather than showing a gradual growth of the field since its introduction in the early 2000s. The number of publications each year may fluctuate depending on funding as most research that includes the development of context-aware FL learning systems or applications tends to be dependent on government policies and funding. For instance, in 2005, the Taiwan Government proposed the ‘Ubiquitous Taiwan’ plan, which promoted a variety of ubiquitous services and technologies and provided support through funding and related research. This may explain the fact that 30 of the 88 total studies and 24 of the 75 empirical studies were conducted in Taiwan or by Taiwanese scholars. Moreover, of the 88 total studies, 41 acknowledged that the research was funded. It is understandable that developing new systems requires funding; however, this tendency of dependence on funding seems to have had a certain effect on the publication patterns and may influence sustainability or future research directions in this field. To increase the sustainability of research in this field, collaborations such as sharing systems across different learning environments or collaborative research between different institutes would be helpful.

Practicality of research and the technology

The use of technology in education requires two important qualities: scalability and sustainability (Shadiev et al., 2017). Scalability refers to the ability to be adapted to diverse contexts with diverse populations, while sustainability is the ability to remain in use. Based on the results of the current study, research on context-aware technology in FL learning seems to lack both scalability and sustainability. Most of the studies focused on developing new applications, but few of these applications have been widely used in different contexts or in different countries. With regard to systems, although a student’s L1 may not have much influence on his or her use of the device, the TL is more sensitive because TL is the system’s input language. However, some systems are not compatible with certain languages, which limit the users of those languages. In actuality, although the studies claimed seamless learning that ‘would not be discontinued due to location changes’ (Shih et al., 2009, p. 375), it is questionable whether those systems or applications are applicable to different languages and in different contexts. Furthermore, to benefit student FL learning, the systems should be able to be utilized over the long term; however, most of the systems appearing in the studies were applied for only a short period. Except for a few studies, most of the studies that involved system development did not engender follow-up studies. Only a few systems, such as HELLO and JAMIOLAS, were later upgraded and re-tested with additional features. Some systems,

such as Mentira, were also tested with different languages or in different situations. However, such follow-up studies are still quite marginal.

Context-aware technology in FL learning requires enhancements to its practicality to allow a single system to be utilized by a wide range of populations and with various TLs in a variety of authentic contexts. Considering that the technology is both context- and user-sensitive, developing a more generically and widely used system despite the user's language and location may be challenging, but if that challenge is met, the results will certainly be rewarding. Additionally, one way to make the technology more sustainable is to make the system more learner-centred and allow students more flexibility. Compared with traditional FL classrooms, learning with context-aware technology provides students with far more freedom and flexibility. However, many of the reported systems were equipped to some degree with specific instructor- or developer-manipulated instructional designs. As shown in the studies (Huizenga, Admiraal, Akkerman, & Dam, 2009; Liu & Chu, 2010; Yang, 2006), when students have more control over their learning and opportunities for their own creation and peer cooperation, they tend to find the activity more interesting and meaningful, which will ultimately lead to the sustainability of learning with the technology.

Conclusion

This study examined the trends and effectiveness of context-aware technology on FL learning based on studies published since 2000. The majority of the selected studies reported that context-aware technology resulted in positive effects on learners' emotions, behaviours, and language outcomes. The results of this study showed that the development and testing of new systems has been quite robust since 2010. However, many of the studies focused primarily on system descriptions and usability tests, and most were limited to one-time trials without follow-ups or more in-depth studies using the same technology. Hence, the trend has been driven primarily by technology rather than pedagogy. This study also addressed problems in research methodologies, research sustainability, and system practicality.

The limitation of the present study is that it was unable to review the specific technologies used in the reviewed studies. Hence, this study could not precisely gauge the functionality of each technology, such as the possibility of alternative input languages or system applicability in different locations and under different conditions when those aspects were not reported in the studies themselves.

In this era of cutting-edge technology, context is not confined merely to the physical space in which activities occur. Instead, it should be considered a dynamic process in which learners' experiences, knowledge, and social interactions are constantly emerging and enmeshing. Cognitive activity should also be viewed as 'the interplay between persons and resources that are distributed across and social and material environments' (Thorne & Hellermann, 2017, p. 722). In this hyper-contextualized situation, learners are actively engaged in the creation and re-creation of context, can generate mixed reality by blending the real world with virtual worlds, and can reconceptualize context (Lee & Park, 2019). In the learner-generated context, learners play the role of producers rather than passive consumers, as in traditional classrooms (Luckin, 2008). Learning with the technology redefines the role of teachers as well. Their role is no longer limited to providing linguistic input and feedback; instead, they can 'help learners curate their personalized learning experiences' (Blyth, 2018, p. 230). Context-aware technology is capable of shifting the FL learning experience from second-hand experiences in classroom environments that place limits on the possible learning experiences to real-world experience with timely and context-adaptive support. Considering the importance of context in FL learning and the potential benefits of context-aware technology, the technology will likely be utilized increasingly frequently. As we continue to develop systems beneficial to FL learning, we also need to focus more on pedagogy. The current study revealed that context-aware technology facilitates the overall FL education through a synthetic review; however, it is not the technology but, rather, the pedagogy that undergirds the use of technology, which determines success in education.

Disclosure statement

No potential conflict of interest was reported by the author.

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